

Glucosinolates in Brassica vegetables: the influence of the food supply chain on intake, bioavailability and human health.

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Glucosinolates (GLSs) are found in Brassica vegetables. Examples of these sources include cabbage, Brussels sprouts, broccoli, cauliflower and various root vegetables (e.g. radish and turnip). A number of epidemiological studies have identified an inverse association between consumption of these vegetables and the risk of colon and rectal cancer. Animal studies have shown changes in enzyme activities and DNA damage resulting from consumption of Brassica vegetables or isothiocyanates, the breakdown products (BDP) of GLSs in the body. Mechanistic studies have begun to identify the ways in which the compounds may exert their protective action but the relevance of these studies to protective effects in the human alimentary tract is as yet unproven. In vitro studies with a number of specific isothiocyanates have suggested mechanisms that might be the basis of their chemoprotective effects. The concentration and composition of the GLSs in different plants, but also within a plant (e.g. in the seeds, roots or leaves), can vary greatly and also changes during plant development. Furthermore, the effects of various factors in the supply chain of Brassica vegetables including breeding, cultivation, storage and processing on intake and bioavailability of GLSs are extensively discussed in this paper.